THERMAL FAULT DIAGNOSTIC REPORT

Automated Infrared Thermographic Analysis & Deep Learning Evaluation

Analysis ID:	a6c5cdea-7ae6-431d-9618-1f4faf60aa28	**Timestamp:**	2026-10-14T13:37:36
Image Filename:	sample_electrical_fault_1.png	**Model Version:**	ResNet18-ThermalTransfer (v1.2.0-trained)
Condition:	**Fault Detected**	**Fault Class:**	**Overheating**
Confidence Score:	53.4%	**Severity Level:**	**Medium**

Thermographic Inspection Imagery

Figure 1: Original Thermal Image

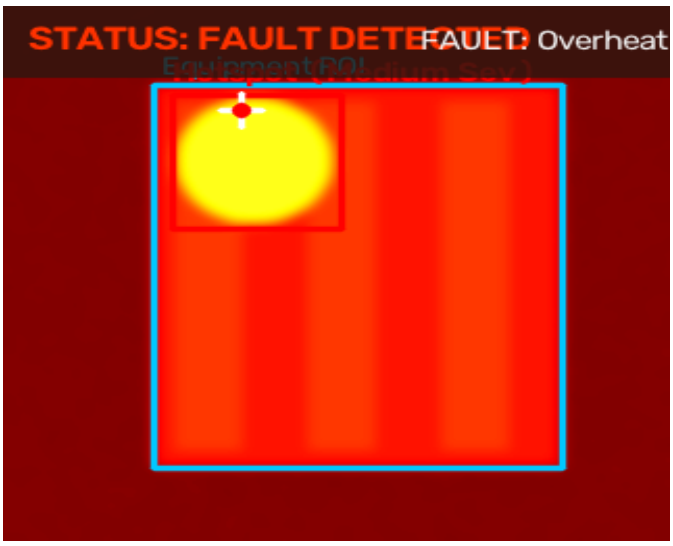

Figure 2: AI Diagnostic Overlay (ROI & Hotspot)

Hotspot Localization & Quantitative Metrics

Hotspot Bounding Box:	X: 81..162, Y: 53..133 (Width: 81px, Height: 80px)
Centroid Coordinates:	[120, 92] px
Peak Thermal Intensity Pixel:	[114, 62] px (Max Intensity: 239.0)
Relative Contrast / Area:	Contrast Ratio: +43.0% | Area Ratio: 11.83%

Engineering Interpretation & Recommendations

Summary: Moderate thermal anomaly identified (Overheating). Local hotspot exceeds equipment operating baseline by 43.0%. Schedule verification within standard maintenance cycle.

Next Actions: Schedule immediate physical inspection of identified terminal contacts. Verify torque on bolted electrical connections, inspect dielectric insulation, and measure phase current balance.

LIMITATIONS & DISCLAIMER:

• Severity levels are derived from thermal gradient contrast and geometry. Uncalibrated digital image formats (PNG/JPG) do not provide calibrated radiometric sensor data in degrees Celsius.

• Solar reflection, airflow, and surface emissivity affect infrared measurements.

• **Automated analysis supports inspection and should not replace qualified maintenance assessment.**